VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY

PROJECT PROGRESS REPORT – NO. 1

Course: Machine Learning
Semester 252

1. GENERAL INFORMATION

Instructor: Dr. Truong Vinh Lan

Report Date: March 08, 2026

Report No.: Progress Report No. 1 / 4

Contact Email: lantv@hcmut.edu.vn

2. GROUP INFORMATION

No.	Full Name	Student ID	Role
1	Nguyen The Hoang	2352354	**Team Leader**
2	Ngo Tran Dinh Duy	2352179	Member
3	Nguyen Viet Nam	2352780	Member
4	Trinh Gia Hiep	2352343	Member
5	Tran Lam Bieu	2352116	Member

3. ASSIGNMENT SCOPE

The group has thoroughly studied the assignment specification and decided to complete all three assignments, aiming for a comprehensive application of traditional ML pipelines across different data modalities. Details are as follows:

Assignment	Data Type	Description
Assignment 1	Tabular Data	Build a traditional ML pipeline on tabular data: EDA, missing value imputation, categorical encoding, feature scaling (MinMax/Standard), dimensionality reduction with PCA, and comparison of Logistic Regression, SVM, Random Forest.
Assignment 2	Text Data	Text classification pipeline: traditional feature extraction (TF-IDF, BoW, n-gram) and modern deep learning embeddings (Word2Vec, BERT). Save embeddings as .npy/.h5 and compare both approaches.
Assignment 3	Image Data	Image classification pipeline: extract features using pretrained models (VGG16, ResNet, ViT), save features as .npy/.h5, and classify using Logistic Regression, SVM, and Random Forest.

4. CURRENT PROGRESS

As of March 08, 2026, the group has completed or is currently working on the following tasks:

No.	Task	Responsible	Status
1	Read and analyze the full assignment specification	Entire team	Completed
2	Register group on the course Google Sheet	Nguyen The Hoang	Completed
3	First team meeting: align on approach and initial task assignment	Entire team	Completed
4	Initialize GitHub repository and set up project folder structure	Nguyen The Hoang	Completed
5	Study theoretical background of traditional ML pipelines, preprocessing and feature extraction techniques	Entire team	In Progress
6	Survey candidate datasets for Assignment 1 – Tabular Data (must have missing values, categorical values, sufficient size)	Nguyen The Hoang Ngo Tran Dinh Duy	In Progress

7	Survey candidate datasets for Assignment 2 – Text Data (diverse labels, sufficient sample size)	Nguyen Viet Nam Trinh Gia Hiep	In Progress
8	Survey candidate datasets for Assignment 3 – Image Data (sufficient size, compatible with Colab environment)	Tran Lam Bieu	In Progress

5. PLAN UNTIL PROGRESS REPORT NO. 2 (22/03/2026)

In the next phase, the group plans to complete the following tasks:

No.	Task	Responsible	Deadline
1	Finalize datasets for all 3 assignments; discuss and confirm with instructor	Nguyen The Hoang (Team Leader)	12/03/2026
2	Download Assignment 1 dataset, perform EDA: descriptive statistics, check missing values and categorical features	Nguyen The Hoang Ngo Tran Dinh Duy	18/03/2026
3	Download Assignment 2 dataset, perform EDA: text length distribution, word frequency, label distribution	Nguyen Viet Nam Trinh Gia Hiep	18/03/2026
4	Download Assignment 3 dataset, perform EDA: label distribution, image size, number of color channels	Tran Lam Bieu	18/03/2026
5	Draft pipeline structure for all 3 assignments and initialize Google Colab notebooks	Entire team	22/03/2026

The group commits to adhering to the schedule and fully meeting all requirements of the assignment. Any updates will be presented in subsequent progress reports.

Ho Chi Minh City, March 08, 2026

Group Representative – Team Leader

Nguyen The Hoang